

Collaborative Git Workflow

Building a Student Information Web Application as a Three-Developer Team

Course	DevOps — MCA, Semester 5
Institution	Christ University, Dept. of Computer Science
Submitted by	Arpan Mukherjee (Team Lead / Developer)
Group	15 — 3 developers
Date	9 September 2026

Repository

<https://github.com/Arpan7125/student-info-app>

1. Learning Objectives

This exercise uses a deliberately simple web application as the vehicle for practising a collaborative Git workflow. The application is not the point; the workflow is. On completion the team can:

- Configure Git locally and verify the configuration
- Create and clone a remote repository on GitHub
- Create, switch between and publish feature branches
- Write meaningful commit messages
- Open, review and merge Pull Requests
- Deliberately create a merge conflict and resolve it
- Work in parallel without overwriting another developer's changes

2. Team and Roles

Student	Role	Responsibility
Arpan Mukherjee 2547116 · @Arpan7125	Team Lead / Developer	Repository creation, base application, reviewing every Pull Request, integration
Joshua George Philip 2547124	UI Developer	HTML structure and CSS styling, contact section, application heading
M A Smera 2547131	JavaScript Developer	Show/Hide details behaviour, application title, conflict resolution

To complete before submission

Both teammates' email addresses and GitHub usernames are still outstanding — in this table, in the application's Project Team cards, and in `README.md`.

3. The Application

A single-page Student Information System built from three static files — no database, no backend, no build step. It displays one student record, reveals additional details on demand, and lists the project team.

File	Purpose
<code>index.html</code>	Page structure — student record, details panel, contact links, team cards
<code>style.css</code>	Design tokens, card components, animated backdrop, light and dark themes
<code>script.js</code>	Details expand/collapse and the pointer-tracked card sheen
<code>README.md</code>	Project documentation and workflow record

Technologies used

- **HTML5** — semantic structure (`<d1>` record rows, inline SVG icons)

- **CSS3** — custom-property design tokens, flexbox and grid, keyframe animation, automatic dark mode via `prefers-color-scheme` , `prefers-reduced-motion` support
- **Vanilla JavaScript** — no libraries, no dependencies
- **Git** 2.55.0 and **GitHub** (repository, Pull Requests, code review)
- **GitHub CLI** 2.100.0, **VS Code**, browser

4. Application Screenshots

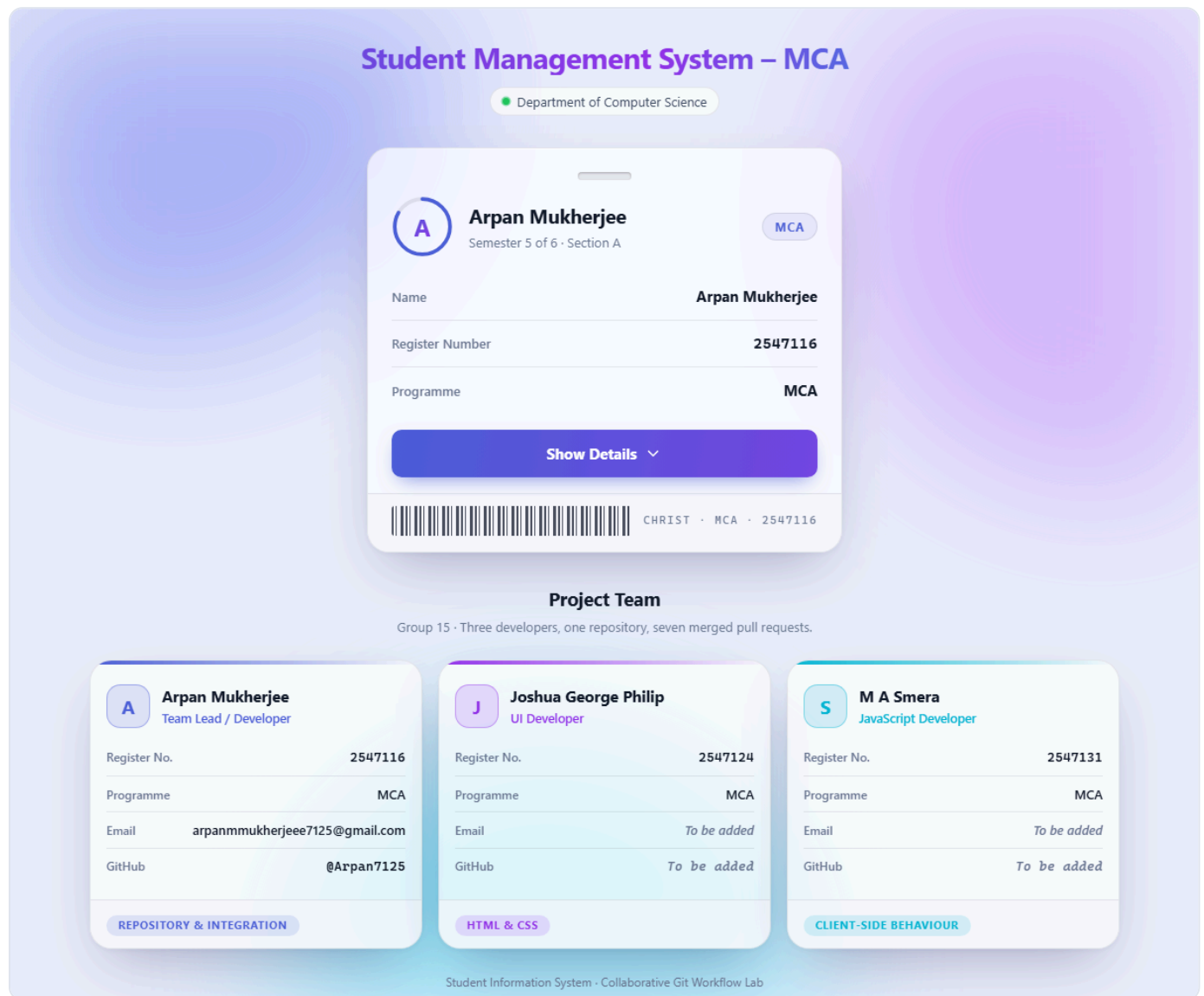


Figure 1 — Default view (light theme). Student record card with semester progress ring and barcode strip, above the three Project Team cards.

Student Management System – MCA

Department of Computer Science

A

Arpan Mukherjee
Semester 5 of 6 · Section A

MCA

Name	Arpan Mukherjee
Register Number	2547116
Programme	MCA

Hide Details ^

Semester	5
Section	A
Mentor	Dr. Cynthia T

CONTACT INFORMATION

Email	arpanmmukherjee7125@gmail.com
Phone	To be added

 CHRIST · MCA · 2547116

Project Team

Group 15 · Three developers, one repository, seven merged pull requests.

A

Arpan Mukherjee
Team Lead / Developer

Register No.	2547116
Programme	MCA
Email	arpanmmukherjee7125@gmail.com
GitHub	@Arpan7125

REPOSITORY & INTEGRATION

J

Joshua George Philip
UI Developer

HTML & CSS

S

M A Smera
JavaScript Developer

CLIENT-SIDE BEHAVIOUR

Figure 2 — After clicking **Show Details**: semester, section and mentor slide open, the button becomes **Hide Details** and the chevron rotates.

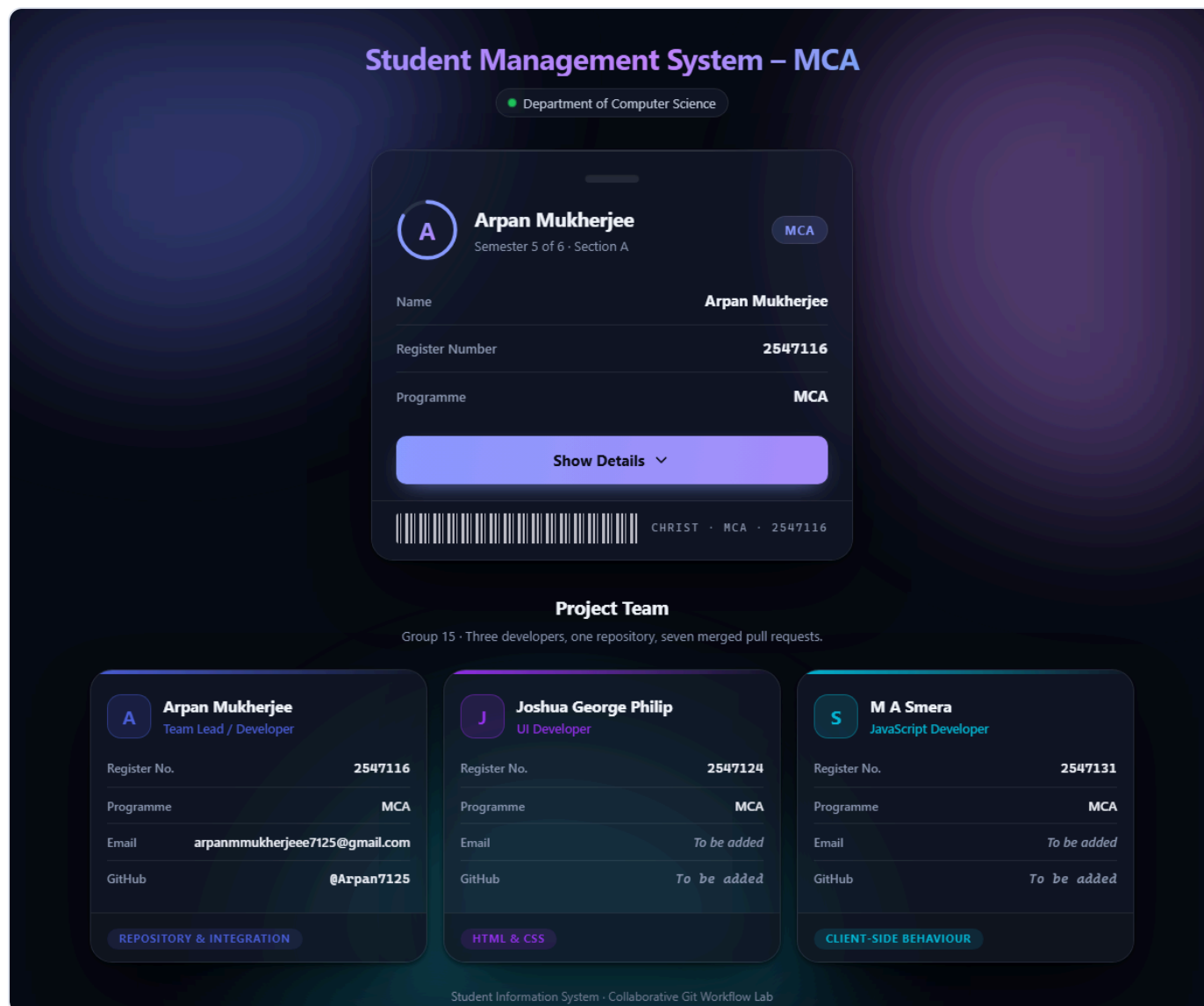


Figure 3 — The same page in dark theme. Every colour is a token redefined under `prefers-color-scheme: dark`; nothing is hardcoded.

Student Management System -

Department of Computer Science



Arpan Mukherjee

Semester 5 of 6 · Section A

Name

Arpan M

Register Number

2

Programme

Show Details ▾



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Project Team

Group 15 · Three developers, one repository, seven merged pu



Arpan Mukherjee

Team Lead / Developer

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arpanmmukherjee7125@

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REPOSITORY & INTEGRATION



Joshua George Philip

UI Developer

Register No.

Programme

Email

GitHub

To

Student Management System -

Department of Computer Science



Arpan Mukherjee

Semester 5 of 6 · Section A

Name

Arpan M

Register Number

2

Programme

Hide Details ▲

Semester

Section

Mentor

Dr. C

CONTACT INFORMATION



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Phone

To



CHRIST · MCA ·

Project Team

Group 15 · Three developers, one repository, seven merged pu



Arpan Mukherjee

Team Lead / Developer

Register No.

Programme

Email

arpanmmukherjee7125@

GitHub

@A



Figure 4 — Mobile (375 px): cards stack to one column, no horizontal scrolling.

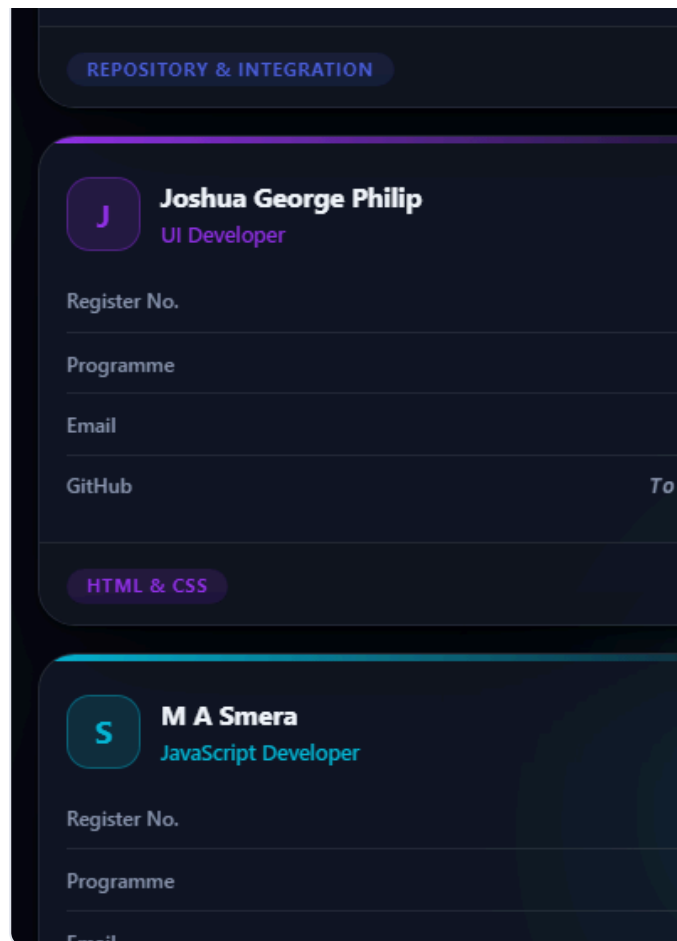


Figure 5 — Mobile, dark theme, details expanded.

5. Git Branching Strategy

`main` is protected by team convention: **nobody commits to it directly**. All work happens on a short-lived feature branch, which is published and merged back through a reviewed Pull Request. Each new branch is created from a freshly pulled `main`.



Branch	Author	Purpose
main	—	Always-working integration branch
feature/ui	Team Member 2	Card layout, spacing, typography, button styling
feature/javascript	Team Member 3	Show/Hide details behaviour
feature/contact	Team Member 2	Contact information section
feature/student-name	Team Member 2	Heading → "Student Management System"
feature/app-title	Team Member 3	Heading → "MCA Student Information Portal"
feature/redesign	Team Member 2	Campus ID-card redesign and Project Team cards
feature/docs	Team Lead	README, lab report and screenshots
feature/team-details	Team Lead	Real team member names and register numbers

Commands used

```
$ git config --global user.name "Arpan Mukherjee"
$ git config --global user.email "arpanmmukherjee7125@gmail.com"
$ git config --global --list

$ gh repo create student-info-app --public
$ git clone https://github.com/Arpan7125/student-info-app.git
$ cd student-info-app

$ git add .
$ git commit -m "Initial student information app"
$ git push -u origin main

# each developer, for each piece of work:
$ git checkout main
$ git pull origin main
$ git switch -c feature/<name>
# ... edit files ...
$ git add .
$ git commit -m "<meaningful message>"
$ git push -u origin feature/<name>
# then open a Pull Request on GitHub for the team lead to review
```

6. Commit History

Every message states what changed. Messages such as `update`, `changes`, `final` or `latest` were avoided.


```

$ git log --oneline --graph

* 7ec5c3c Merge pull request #9 from Arpan7125/feature/register-numbers
|\
| * b2bb2fb Correct the team register numbers
|/
* 31d3b9d Merge pull request #8 from Arpan7125/feature/team-details
|\
| * 6840063 Add real team member names and register numbers
|/
* eacab65 Merge pull request #7 from Arpan7125/feature/docs
|\
| * ee598ae Add project documentation, lab report and screenshots
|/
* 9b8119e Merge pull request #6 from Arpan7125/feature/redesign
|\
| * 914864e Redesign the interface as an animated campus ID card
|/
* b6ea35a Merge pull request #5 from Arpan7125/feature/app-title
|\
| * e65885f Resolve merge conflict in application title
| |\
| |/\
|/|
* | f957fab Merge pull request #4 from Arpan7125/feature/student-name
|\ \
| * | 17cf9f8 Update application heading
|/ /
| * c9c926f Update application title
|/
* bf45ea7 Merge pull request #3 from Arpan7125/feature/contact
|\
| * d72f9ae Add contact information
|/
* 9599709 Merge pull request #2 from Arpan7125/feature/javascript
|\
| * de46f44 Add student details functionality
* | 2dca70b Merge pull request #1 from Arpan7125/feature/ui
|\ \
| |/\
|/|
| * ed57371 Improve student information UI
|/
* 2679d23 Initial student information app

```

The graph reads bottom-up. Each `|\` fork is a feature branch leaving `main`; each merge commit is a Pull Request being accepted.

7. Pull Requests

PR	Branch → main	Author	Status
#1	feature/ui	Team Member 2	MERGED
#2	feature/javascript	Team Member 3	MERGED
#3	feature/contact	Team Member 2	MERGED
#4	feature/student-name	Team Member 2	MERGED
#5	feature/app-title	Team Member 3	MERGED after conflict resolution
#6	feature/redesign	Team Member 2	MERGED
#7	feature/docs	Team Lead	MERGED
#8	feature/team-details	Team Lead	MERGED
#9	feature/register-numbers	Team Lead	MERGED
#10	feature/conflict-evidence	Team Lead	MERGED

Review checklist applied by the team lead

1. Does the application still work in the browser?
2. Is the code understandable to someone who did not write it?
3. Is the change relevant to the branch's stated purpose?
4. Is the commit message meaningful?

8. The Merge Conflict

8.1 What caused it

Branches `feature/student-name` and `feature/app-title` were both created from the **same commit** on `main` (`bf45ea7`), and both edited **the same line** of `index.html` :

Branch	Line 11 of index.html
<code>feature/student-name</code>	<code><h1>Student Management System</h1></code>
<code>feature/app-title</code>	<code><h1>MCA Student Information Portal</h1></code>

PR #4 was merged first, so `main` moved forward. When PR #5 was then merged, Git had two different changes to the same line sharing one common ancestor, and no rule for deciding which is correct — so it stopped and asked a human.

```
$ git merge main
Auto-merging index.html
CONFLICT (content): Merge conflict in index.html
Automatic merge failed; fix conflicts and then commit the result.

$ git status --short
UU index.html
```

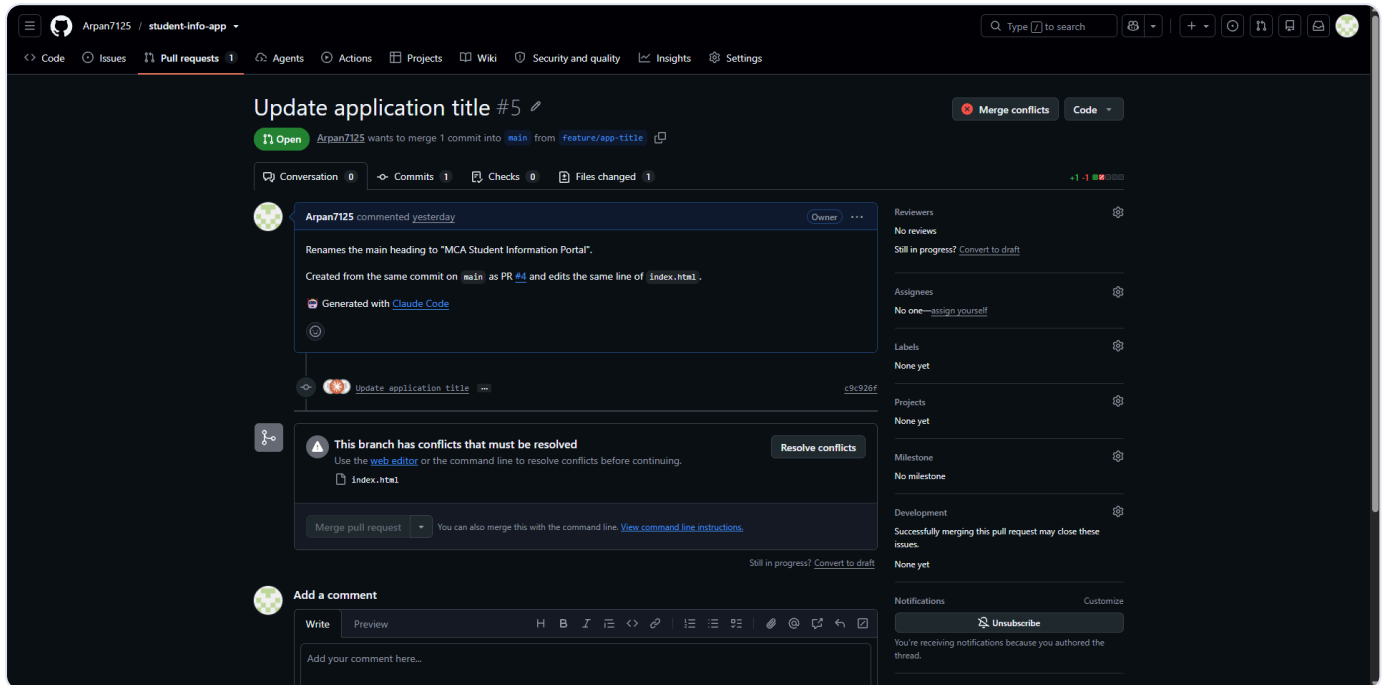


Figure 6 — Pull Request #5 blocked on GitHub. The red **Merge conflicts** badge sits beside the title, the banner reads “*This branch has conflicts that must be resolved*” naming `index.html`, and **Merge pull request** is disabled until a human resolves it.

8.2 The conflict markers

```
<<<<<<< HEAD
    <h1>MCA Student Information Portal</h1>
=====
    <h1>Student Management System</h1>
>>>>>> main
```

Marker	Meaning
<<<<<<< HEAD	Start of the version on <i>your current branch</i>
=====	Divider between the two versions
>>>>>> main	End of the version coming in <i>from main</i>

8.3 How it was resolved

Rather than discarding either teammate's intention, the two were combined into a single heading. All three marker lines were deleted.

```
$ git checkout main
$ git pull origin main
$ git checkout feature/app-title
$ git merge main          # conflict reported here

# edit index.html - keep the combined heading, remove every marker

$ git add index.html
$ git commit -m "Resolve merge conflict in application title"
$ git push origin feature/app-title
```

Final line in `index.html` :

```
<h1>Student Management System - MCA</h1>
```

The Pull Request then became mergeable and the team lead merged it.

Key takeaway

A merge conflict is not an error and not a failure. It is Git refusing to guess when two developers disagree about the same line — and correctly handing the decision back to the team.

9. How to Run the Application

No server, no dependencies and no build step are required.

```
$ git clone https://github.com/Arpan7125/student-info-app.git
$ cd student-info-app
```

Then open `index.html` in any browser — double-click it, or:

```
start index.html    # Windows
open index.html     # macOS
xdg-open index.html  # Linux
```

Click **Show Details** to expand the additional student information. The page follows the operating system's light or dark theme automatically.

10. Deliverables Checklist

Deliverable	Required	Status	Notes
Repository with 4 files	4	DONE	index.html, style.css, script.js, README.md
Branches	6	DONE	11 on the remote: main + 10 feature branches
Meaningful commits	≥ 5	DONE	15 commits, all descriptive
Pull Requests	≥ 3	DONE	10 opened, all 10 merged
Conflict evidence	Screenshots	DONE	GitHub conflict screenshot (Figure 6) and terminal capture
README with all sections	Yes	DONE	In the repository
Real teammate names	Yes	DONE	Arpan Mukherjee, Joshua George Philip, M A Smera
Register numbers	Yes	DONE	Arpan 2547116, Joshua 2547124, Smera 2547131